Position-Based Flocking for Persistent Alignment without Velocity Sensing

1st Kasper Bloch Mathiasen
Drones and Autonomous Systems
SDU UAS
Odense, Denmark
kmath22@student.sdu.dk

2nd Aksel Møller-Hansen
Drones and Autonomous Systems
SDU UAS
Odense, Denmark
akmo22@student.sdu.dk

3rd Harald Bay Baastrup
Drones and Autonomous Systems
SDU UAS
Odense, Denmark
hniel22@student.sdu.dk

***Abstract*—This project investigates position-based flocking for persistent alignment without direct velocity sensing implemented by Jond et al. [5]. Three flocking models were implemented and compared in a 2D agent simulation: velocity-based alignment, position-based alignment, and position-based alignment with a threshold mechanism. The models were evaluated using average flock speed and a global alignment metric across multiple simulation runs. Results show that the threshold-based position model achieves persistent alignment comparable to the velocity-based approach, while avoiding direct velocity sensing. This makes the method more suitable for real-world robotic swarm systems as true velocity sensing is not feasible.**

I. Introduction

Biological flocking demonstrates how animal groups can show complex behaviour, that is synchronized to the point where it appears as the individual animals are centrally controlled [1]. However, biological flock motion emerges from actions of individual animals using their local perception rather than through some global controller [2]. This can be seen in large bird flocks where each bird responds to nearby neighbouring birds, which still results in an collective coherent flocking behaviour. The contrast between complex behaviour of flocking and the simple behaviour of individual animals is the foundation for using biological flocking as inspiration [2]. Prior to implementing flocking behaviour in actual robotic application, simulation is commonly used, where robots are simplified to agents for implementation of new methods [4].

For implementation of autonomous flocking systems, a centralized control approach will for a limited number of agents be feasible [3]. However, challenges will occur when scaling the number of agents, where requirements for communication and computations becomes demanding at large scale [11] [12]. These limitations are often less visible in simulated environments. This motivates using biological flocking as inspiration [1], where agents will be limited to local knowledge of nearby agents' position instead of relying on global knowledge of all agents' position.

Implementation of flocking behaviour on physical robots like Unmanned Aerial Vehicles (UAV) introduces sensing limitations, that are often hidden in simulation. When working with flocking, the measurement of velocity of neighbouring agents can be difficult [5], while relative positional information is more accessible through sensors like lidar or cameras. If agents could act alone based on locally available sensor measurement, the need for communications in-between all agents will be reduced.

In this study, we reproduce the work of Jond et al. [5], in which position-based sensing is employed for alignment instead of velocity-based sensing. This approach is relevant because it explores how alignment can emerge without directly measuring neighbouring agents' velocity. This makes the model suggested more suitable under more realistic scenarios with sensing and communication limitations. The aim of this project is therefore to reproduce the position-based flocking behaviour and evaluate how well the swarm maintains a common alignment and cohesion when relying on position-based sensing.

II. Related Works

A foundational contribution to distributed behavioural modelling in bio-inspired swarm robotics is the work of Reynolds [2]. In Reynolds' boids model, collective motion emerges from decentralized agents that operate using only local information and follow three interaction rules: separation, alignment, and cohesion. Together, these rules give rise to complex global behaviour in multi-agent systems that can collectively exhibit coherent motion patterns.

The work by Bhattacharyya et al. [8] builds on Reynolds' model [2], but argue that reducing the number of parameters is possible, while still preserving flocking behaviour. In their formulation, the classical boids model consists of separation, alignment, cohesion and neighbourhood vision range, which is four different tunable parameters. By removing the vision radius and combining cohesion and separation, they are left with two parameters (alignment and cohesion-separation), which reduces the complexity to two tunable parameters. This is similar to the paper by Jond et al. in [5], which also combines cohesion and separation into one parameter. Another related simplification is presented by Souza et al. [10], that propose to use a steering control law, and a velocity control law, that also will reduce the number of parameters to two.

In cases where intensive communication and velocity sensing is possible, Zhang et al. [7] proved that it is possible to align a flock using a different approach. It is based on situations inspired by real-world scenarios such as traffic jam motion and escaping with panic behaviour. This shows, that research in swarming uses different approaches to modelling the biologically inspired flocking behaviours, as this paper introduces traffic jams and escaping with panic, which is inspired from human behaviours in different situations.

Kahani et al. [9] address the robustness of flocking systems in the presence of agent escaping their flocks. Extending Reynolds' three-rule framework, they propose an algorithm in which agents in a flock adapt and follow a rogue agent based on the velocity at the time of an agent's departure and the number of neighbouring agents. This approach enhances the system's ability to maintain cohesion despite disruptions.

Another relevant application is the study by Cheraghi et al. [4], which investigates coordinated motion and interaction in swarm networks based on individualized and local algorithms. In this study, each agent is able to interact with both other neighbouring agents and passive entities in the environment. The authors introduce both two-dimensional and three-dimensional simulated environments in which agents can communicate and navigate through pheromone-based signalling, environmental markings, and through direct inter-agent interactions. This local sensing approach aligns well with the position-based alignment introduced by Jond et al. [5], where each agent responds to neighbouring agents rather than relying on global knowledge of all agents.

Murakami et al. [6] propose an alternative approach for coordinated swarm motion. They propose that coherent behaviour can emerge by mutual expectations, rather than directly measuring and matching velocities between agents. Here agents act based on predicted future states of neighbouring agents instead of their current headings, which allows coordinated swarm motion to arise without the traditional explicit alignment rules. Their study suggests that emergent coordination stems from more implicit interactions, rather than relying on explicit velocity measurement not applicable in real scenarios. The idea of being independent of velocity measurements and global knowledge of all agents aligns well with the approach introduced by Jond et al. [5], since both works suggest that flocking can be achieved using local interaction rules without relying on information that may be difficult to obtain directly in robotic swarms.

Together, these works illustrate how coordinated swarm behaviour can emerge from individual interaction with neighbouring agents instead of depending on velocity-based alignment, which altogether shows the relevance of the position-based alignment approach proposed by Jond et al. [5].

III. Methods

A. Velocity-based alignment

Like in the paper by Jond et al. [5], we implement a velocity-based flocking model for alignment as a baseline. The total number of agents are given by $N \in \{1, ..., n\}$, where each agent is indexed by $i \in N$. This notation will be used throughout the equations describing the agents. The individual agents are described by an position, velocity and acceleration, which will be used to control the agents. The baseline control input used for calculating acceleration is defined in equation (1), that calculates the acceleration of for each agent [5]. Here the position vector is denoted $\vec{p}$ and the velocity vector denoted $\vec{v}$, where the index j denotes the neighbouring agents to the current agent indexed i, that the control input $\vec{u_i}$ is calculated for. Where ψ is the cohesion-seperation gain which will be explained below.

$$\vec{u_i} = \sum_{j \in N_i} \psi(\|\vec{p_j} - \vec{p_i}\|)(\vec{p_j} - \vec{p_i}) + \sum_{j \in N_i} (\vec{v_j} - \vec{v_i}) \quad (1)$$

The first sum is the cohesion-separation, and the second sum velocity alignment. This model establishes a baseline as it balances cohesion and separation, so the model should repel neighbours as the cohesion increases, and attract them when separation increases. Meanwhile, the alignment part will ensure, that the agents moving in a flock will align their speeds by reducing the velocity differences between neighbouring agents. This specific alignment method in equation (1) is referred to as velocity-based alignment throughout this paper.

The cohesion-separation gain is described in equation (2), which regulates the balance between cohesion and separation depending on its parameter δ_i.

$$\psi(||\vec{p_j} - \vec{p_i}||) = 1 - \frac{\delta_i |N_i|}{\|\vec{p_j} - \vec{p_i}\|} \quad (2)$$

This gain creates a distance-dependent interaction with an equilibrium point, when the distance between an agent and its neighbours is equal to the parameter δ_i times the number of neighbours $|N_i|$, as shown in equation (3).

$$\|\vec{p_j} - \vec{p_i}\| = \delta_i |N_i| \quad (3)$$

At this equilibrium point, an agent will neither be repulsed from- or attracted towards its neighbours, as the agent is at an equilibrium state without cohesion or separation between the agents. The value of the parameter δ_i times the number of neighbours $|N_i|$ is crucial for an agents cohesion-separation behaviour. In the case that the distance between an agents and its neighbours is above this value ($\delta_i |N_i|$), the resulting cohesion-separation gain becomes larger than zero, $\psi > 0$, which results in cohesion between agents. Contrary, if the distance becomes smaller than value ($\delta_i |N_i|$), the resulting cohesion-separation gain becomes smaller than zero, $\psi < 0$, which results in a separation between agents. The parameter δ_i is used to adjust the desired spacing generally, while the size of each flock $|N_i|$ determines the spacing for the specific flock. Thus, the cohesion-separation gain ψ is essential to ensure that the flock maintains a stable formation by preventing flocks from collapsing into a single unit while still preserving a sufficient separation between agents.

After the control input $u_i(t)$ have been calculated, the acceleration is integrated over a time step Δt to update the velocity of each agents as shown in equation (4).

$$v_i(t + \Delta t) = v_i(t) + \Delta t u_i(t) \quad (4)$$

After this update, the speed of the agents is limited by a maximum speed by scaling down the velocity while ensuring the same direction as shown in equation (5).

$$\vec{v_i} = \begin{cases} \vec{v_i} & \text{if } \|\vec{v_i}\| \leq \vec{v}_{max} \\ \vec{v}_{max} \frac{\vec{v_i}}{\|\vec{v_i}\|} & \text{if } \|\vec{v_i}\| > \vec{v}_{max} \end{cases} \tag{5}$$

The purpose of the speed limitation is to make the simulation more realistic, since both animals and physical robot systems have finite speed capabilities. The speed limiter will therefore help ensure stable execution of experiments.

B. Position-based alignment

Due to the nature of animals not being able to measure velocity [5], but rather the change of position over time, position based alignment is a way to mimic biological swarms.

With position based flocking, we approximate the relative velocity using equation (6).

$$\vec{v_j} - \vec{v_i} \approx \frac{(\vec{p_j} - \vec{p_i}) - (\vec{p_j}(0) - \vec{p_i}(0))}{t}, \quad t > 0 \tag{6}$$

This method compares the current relative position between agent *i* and its neighbour *j*, $(\vec{p_j} - \vec{p_i})$, with their initial relative position, $(\vec{p_j}(0) - \vec{p_i}(0))$. The change in relative position is then divided by the elapsed time *t*, which is used to approximate the relative velocity between the two agents. As the elapsed time *t* is the same for all neighbours, the constant $\frac{1}{t}$ can be moved out from the summation over all neighbours for agent *i*.

$$\sum_{j \in N_i} (\vec{v_j} - \vec{v_i}) \approx \frac{1}{t} \sum_{j \in N_i} [(\vec{p_j} - \vec{p_i}) - (\vec{p_j}(0) - \vec{p_i}(0))] \tag{7}$$

This approximation of the velocity using position in equation (7) is close to the velocity-based equation in equation (1), which will be used to provide the control input $\vec{u_i}$ for the agents. By using approach for approximating the relative velocity between an agent and its neighbours, the alignment term in the control input $\vec{u_i}$ can be based purely on positional measurements.

C. Position-based alignment with threshold

From equation (7), it can be seen that when $t \rightarrow \infty$, the position-based approximation of the relative velocity will approach zero in equation (7). In practice, this means that the alignment term is removed gradually from the control input $\vec{u_i}$ as the elapsed time increases.

Like in the paper by Jond et al. [5], we have also implemented a threshold that replaces the decaying term $\frac{1}{t}$ with $\frac{1}{k}$ when $t > \frac{1}{k}$. In the period before the threshold, the flock still has its biological tendency but it also ensures the alignment will persist as the elapsed time increases.

$$\phi = \begin{cases} \frac{|N_i|}{t} & \text{if } 0 < t \leq \frac{1}{k} \\ k|N_i| & \text{if } t > \frac{1}{k} \end{cases} \tag{8}$$

This concludes the three implemented flocking models: velocity-based, position-based, and position-based with threshold.

IV. Environment and Simulation

To test the three different flocking models a simple 2D simulation has been setup. To visualize, OpenCV was used to display the agents during simulations.

When instantiating each agent, they are given a random uniform position and velocity. Their positions are set to be close to the center of the world, but spread out so they do not spawn on top of each other. The same is done for the velocity, where each agent is given a speed within a minimum and maximum range. This speed limit ensures reasonable velocities for a simulation of biologically inspired flocks, where the agents can align their movement.

The simulation domain is limited to be within a finite space, and for that reason the simulation space is made as a toroidal, meaning when a agent exits to one side it is entering from the opposite side. By making the agents wrap around the world, a new problem occurs. When calculating the distance between the agents in a flock while accounting for a toroidal environment, the result is skewed as the positions are now further away. To avoid this issue, an additional function is introduced that handles the calculation of distances in a toroidal world. This allows equations (1) and (7), to be computed, as they both rely on the difference in position.

Just like explained by [8] and [5], we have also implemented a visual radius to enable each agent to do local sensing of the neighbouring agents. Throughout the tests, this radius was kept constant to focus on the three models instead, but could potentially be tuned to increase the efficiency of the models.

To run a test of one model, the python library argparse has been utilized to increase testing throughput, where the parameters shown in table I.

TABLE I
THE APPLICABLE ARGUMENTS FOR DOING A TEST.

Argument	**Information**
–agent	Number of agents
–steps	Number of simulation steps
–max_speed	Number of runs
–runs	Seed for replication
–cv2	Enable or disable OpenCV
–height	Window height
–width	Window width
–mode	Velocity, position, position_threshold

To ensure that the runs are comparable across the different alignment modes, the random seed was manually configured and incremented after each run. This ensures that the agents initialize with the same position and velocity vectors, making the tests replicable.

A. Simulation Setup

The three experiments were carried out with 50 agents in a 700 × 700 m environment with a maximum speed of 2 m/s and a simulation duration of 100 s for each run. Each of the three models was evaluated over 200 runs.

The three alignment models were tested separately with an initial random seed of the value 2, which increments subsequently for each run in an experiment. This ensures that the results were comparable, as the agents were initialized with the same positions and velocities across the different alignment models.

For all three experiments, OpenCV visualization was disabled to improve execution speed.

V. Results

A. Alignment Score γ

To compare our results with those presented in the original work by Jond et al. [5] for the three models we have implemented, we adopt the same parameter settings. To get a better estimate of the implemented models, we ran multiple runs of each of the three models, and calculated the mean value of the all agents metrics at each time step in the simulation. This produces a more stable and representative curve of the system's behaviour, as it removes the impact of the initial random position and velocity, as well as any single, simulation-specific behaviour that diverges from the others. The results are displayed as plots that illustrate both the mean speed and the alignment metric γ averaged across all runs. The mean speed was obtained by averaging the speeds of all agents. The γ alignment score, defined in equation (9), provides a more comprehensive measure of directional alignment of all agents. Its values lie in the interval $\gamma \in [-1, 1]$, where 1 indicates strong directional alignment, 0 corresponds to randomly oriented directions, and -1 reflects perfectly opposed directions..

$$\gamma = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{|N_i|} \sum_{j \in N_i} \frac{\vec{v_i}^T \vec{v_j}}{\|\vec{v_i}\| \|\vec{v_j}\|} \tag{9}$$

The inner term computes the average cosine similarity, which measures how aligned the velocity vectors are between agent i and its neighbours j. The inner term is then summed and averaged, which gives a local alignment score for each agent in relation to its neighbours.
The outer term then sums the local alignment scores and averages over all agents, which in return gives the global directional alignment term γ of the entire flock.

B. Comparison of Alignment Models

At the beginning of the simulation the agents move independently, which is reflected by the alignment score initially being close to zero as seen in fig. 1a. As the agents begin to form a flock, their velocity vectors increasingly align, which is seen in fig 1a, where the alignment score describing the velocity-based model approaches 1.

In both fig. 1a and fig. 1b, the position-based alignment model without threshold quickly loses its alignment because the alignment gain $\frac{1}{t}$ decays over time. This results in the whole flock staying in the same area but mainly performing small cohesion-separation adjustments rather than maintaining an aligned motion in one direction. This also explains why the position-based model without threshold average speed appears more noisy compared to the position-based model with threshold. Once the alignment gain becomes weak, the control

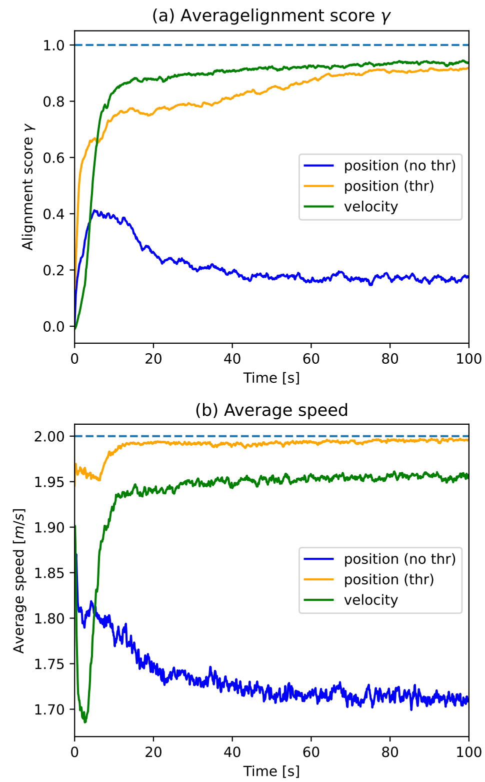

Fig. 1. Comparison of the three alignment methods, where the position-based alignment with threshold uses the alignment term $\frac{|N_i|}{t}$ before the threshold changes. The mean over 200 simulation runs of 100 seconds each, with $k = 0.15$. Subplot (a) shows the average alignment metric γ, and subplot (b) shows the average speed.

input is primarily affected by cohesion-separation adjustments. In contrast, the position-based model with threshold prevents the alignment gain from vanishing, and the average speed converges to an almost constant speed.

However, to compare the position-based model with threshold more directly to the position-based model without threshold, we changed the pre-threshold alignment gain $\frac{|N_i|}{t}$ to $\frac{1}{t}$. This isolates the effect of the threshold by making the initial alignment gain identical between the two position-based models, which can be seen in fig 2. This test was done using the same random seeds as in fig. 1.

The plot in fig. 2 using the alignment gain $\frac{1}{t}$ shows a spike as it surpasses the threshold, which will be addressed below in the discussion.

VI. Discussion

In the original paper by Jond et al. [5], their plot of the average speed shows that the position-based model with threshold mirrors the behaviour of the position-based model without threshold up to around 7 seconds, where the threshold is activated. This is different from what we observe in our implementation: in fig. 1b, where the position-based models have different average speeds before the threshold becomes active. We discovered, that by changing the alignment gain in

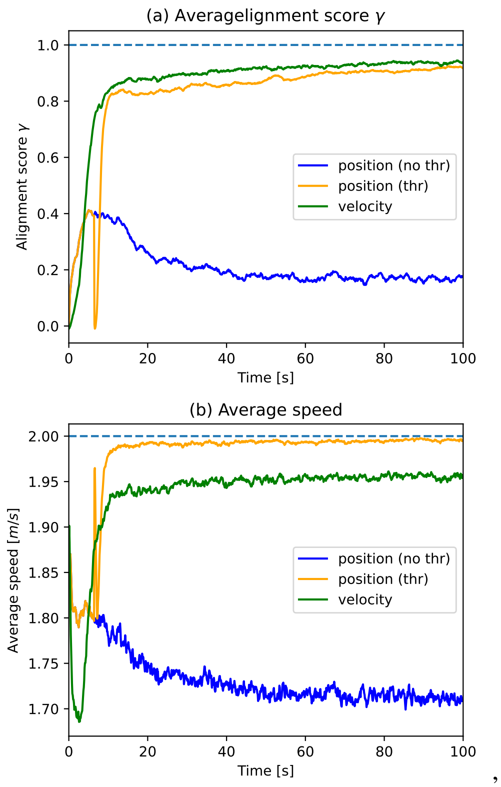

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Fig. 2. Comparison of the three alignment methods, where the position-based alignment with threshold uses the alignment term $\frac{1}{t}$ before the threshold changes. The mean over 200 simulation runs of 100 seconds each, with $k = 0.15$. Subplot (a) shows the average alignment metric γ, and subplot (b) shows the average speed.

the position-based model with threshold from $\frac{|N_i|}{t}$ to $\frac{1}{t}$, the average speed of the two position-based models aligned prior to the threshold activating, which is seen in fig. 2b. One reason for this difference between their plot and our in fig. 1a could be their parameters differs from ours, as they only revealed the size of the value k. If their visual radius were smaller than the one we implemented, the expected number of neighbours within range could be approximately equal to the value 1, which would correspond to $\frac{1}{t}$.

However, a consequence of changing the alignment score is the large spikes seen in fig 2a and 2b. The large spike in average speed seen in fig 2b for the position-based model with threshold, which is after the threshold function is activated. A similar spike is shown in fig 2a, where the alignment score γ decreases for the same position-based model with threshold. The reason for these spikes is that the threshold causes an abrupt change in the alignment gain. Since the gain before and after the threshold activation is different, the agents are affected by a sudden change in control input, which causes them to change their directions rapidly. This directional change will briefly reduce the alignment score γ, which is reflected by the downward spike in fig 2a. And at the same time, the change in motion causes an increase in the agents' average speed, which is reflected as the upward spike in fig 2b. After this short change, the agents adapt to the new alignment gain and the flock stabilizes again, which makes both average speed and the alignment score return to a steady behaviour.

fig accordance with the original paper by Jond et al. [5], fig 2a shows that the reference velocity–based model rapidly converges to a high alignment score. Fig 2a also illustrates that the position-based model with threshold initially mirrors the position-based model without threshold until the point where the threshold function is triggered. Once the threshold functionality is activated, the position-based model with threshold closely tracks the velocity-based model. This supports the original goal of approximating the relative velocity of neighbouring agents by only using a position-based sensing approach.

VII. Conclusion

This project demonstrates how flocking alignment can be achieved using position rather than velocity. This approach, proposed by Jond et al. [5], reduces the need for excessive communication or velocity sensing, and makes the simulated flock more relevant for real-world scenarios. The three different models have been simulated using simple agents to form a flock and evaluated using the same metrics as the original paper.

As in the original paper, the approach of estimating the relative velocity change based on the initial and current position of the agents was found to be feasible. This is supported by comparing both the alignment score γ from our implementation to the original paper, where the same general pattern is observed. The differences in metrics compared to the original paper are likely due to the parameters used in our implementation differing slightly from the original paper. The position-based model without threshold did not obtain a high alignment score over time. However, the implementation of the position-based model with threshold was able to obtain a high alignment score.

We observed, that prior to the threshold activation, the position-based model with threshold did not behave similarly to the position-based model without threshold when using the alignment gain $\frac{|N_i|}{t}$. By modifying the pre-threshold gain to $\frac{1}{t}$, the two position-based models became identical before the threshold was activated.

Overall, the results support that position-based flocking can be used to obtain alignment without direct velocity sensing. We find this positive, since the position-based approaches is more realistic for physical robot systems, where relative position can be measured using sensors such as lidar or cameras instead of direct velocity sensing, which is more difficult to measure.

GitHub Repository

Link to our GitHub repository:
https://github.com/TheJoboReal/bio_swarm_project

LLM USE

Throughout this rapport, LLMs (ChatGPT) has been used to refine sentences, correct grammatical errors, and draft the abstract.

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